

BARRETT RUTH

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EDUCATION

University of Virginia

Bachelor's of Computer Science

Charlottesville, VA

Aug. 2022–May 2026

- Coursework: Advanced Algorithms, Operating Systems, Algorithmic Economics, Group Theory GPA: 3.96/4.00
- Activities: ICPC, HooHacks 2x Winner

EXPERIENCE

Neovim

Core Maintainer

May 2026–Present

Remote

- One of 27 maintainers of the world's most admired editor (3M+ users)
- Added Neovim-native directory browser, replacing the legacy Vim listing
- Implementing git diff and patch highlighting via tree-sitter

Ramp

Backend Software Engineering Intern

September–December 2025

New York City, NY

- Built bulk fraud dispute processor, saving $\approx$ \$100,000/yr and $\approx$ 10,000 CX hours/yr
- Stabilized $\approx$ 10% of backend tests by removing SQLAlchemy session-sharing in Temporal workflows

DRW

Software Developer Intern

June–August 2025

Houston, TX

- Optimized office-wide exposure tooling, permitting 70x increase in load and eliminating production OOMs
- Built “Exchange Equivalence,” a portfolio risk hedging tool for FTR trade exposures
- Ported power-grid topology spanning tree to a Rust/Julia FFI, cutting AWS costs $\approx$ \$10,000/mo

UVA Insight Computer Architecture Lab

Hardware Profiling Research Assistant

January–May 2025

Charlottesville, VA

- Profiled heterogeneous memory systems of NVIDIA GH200 Superchip for large-scale GEMM

Visa

Software Engineering Intern

May–August 2024

Austin, TX

- Built “GenAI Support Assist,” an LLM (GPT) + Ada-002 retrieval pipeline, cutting debugging $\approx$ 8 hrs/wk
- Only intern to deploy to production (CloudView); selected by the SVP of Payment Solutions

PROJECTS

tmux | *C, libevent*

2026–Present

- Added auto-refreshing copy mode, constant-time refresh recloning the viewport (0.015ms average)
- Authored 12 more patches: NULL-deref, leak, double-free, and input-sanitization fixes

Neovim Tooling | *Lua, libuv*

May 2025–Present

- Core maintainer of none-ls, Neovim's largest LSP integration framework (2.9k+ stars, 10k users)
- Built guard, Neovim's first async formatting and linting plugin (6k users) via Lua coroutines

Cavalier Autonomous Racing | *ROS2, C++, Qt*

April 2024–May 2026

- Refactored ROS2 state machine for emergency racing scenarios with in-house YAML validator
- Migrated GUI to Qt C++, increasing telemetry throughput 25% via async monitoring

theCourseForum | *Django, PostgreSQL, AWS*

August 2022–May 2025

- Led largest university course analytics platform serving 20,000+ students (85% of student body)
- Replaced Elasticsearch with Postgres trigram search, removing a service dependency

TECHNICAL SKILLS

Languages: C, C++, Rust, Python, Lua, TypeScript, JavaScript, SQL, Java, Nix, Shell

Technologies/Frameworks: Linux, Docker, Nix, CI/CD, Git, Temporal, PostgreSQL, Redis, Protobuf, AWS, GCP, Node.js, React